

Module 7

Technical Workouts

1. 1. Understand Data Modeling

- a. Entity identification
- b. Relationships
- c. Data modeling principles
- d. Real-world schema design
- e. Scalable database structures

2. 2. Understand Database Normalization

- a. Why normalization matters
- b. First Normal Form
- c. Second Normal Form
- d. Third Normal Form
- e. Avoiding data redundancy

3. 3. Understand SQL Joins

- a. INNER JOIN
- b. LEFT JOIN
- c. RIGHT JOIN
- d. FULL JOIN
- e. Multi-table queries

4. 4. Understand Advanced Querying

- a. Subqueries
- b. Nested queries
- c. Common Table Expressions (CTEs)
- d. Complex filtering
- e. Analytical queries

5. 5. Understand Window Functions

- a. Ranking
- b. Partitioning
- c. Running totals
- d. Analytical calculations
- e. Business reporting use cases

6. 6. Understand Database Performance

- a. Indexes
- b. Query optimization
- c. Execution awareness
- d. Efficient querying
- e. Scalable data retrieval

7. 7. Understand Data Warehousing Concepts

- a. OLTP vs OLAP
- b. Data warehouses
- c. Analytical databases
- d. Reporting systems
- e. ML data pipelines